

The weighted-jet programme: a synthesis (germbij §7.4 inductive recovery, four tides)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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Abstract

Four tides carried the germbij note’s §7.4(a) inductive recovery from scoping to a sorry-free comparison theorem: two finite jets $a x^{2k} + \sum_{r=1}^R c_r x^{2k+r}$ with uniformly positive profiles whose Gibbs moments at the matching observables x^{2k+r} eventually agree have equal coefficients (`polynomialJet_recovery`, PRs #53–#56). The programme was scoped against three alternatives, rerouted once mid-course — the pairwise-difference argument replacing the planned multi-index expansion machinery at roughly half the formal cost — and survived one process failure worth remembering: a merged PR whose build gate had been silently vacuous. This synthesis records the mathematics, the two consults that shaped it, and the incident.

1 What was proven

The headline chain, all in `Laplace/OneD/`:

- **MonomialVariance** (PR #53): Gamma closed forms and strict positivity of $\text{Var}(x^n)$ against the reference weight $e^{-t x^{2k}/(2k)!}$ — the injectivity input, formalising the note’s own “ $\text{Var}[Q] = 0$ forces Q constant” argument via the open nonempty support of the centered integrand.
- **JetScaling** (PR #54): the jet potential factors through its profile, $L_q(u) = u^{2k} P(qu)$, so the correct integrability envelope is a uniformly positive profile ($\rho \leq P$ on all of $\mathbb{R}$), giving $\rho u^{2k} \leq L_q(u)$ for *every* real q ; the exact scaling identity $\langle x^s \rangle_t = q^s \cdot$ (normalized jet moment) under $t q^{2k} = 1$; and the $R = 2$ discriminant certificate $c_1^2 < 4ac_2$, admitting unrestricted odd c_1 .
- **JetDifference** (PR #55, repaired in #56): for jets agreeing below rung r , the difference factors as $L_q^1 - L_q^2 = q^r g(q)$ with $g(0) = \delta_r u^{2k+r}$; the exponential secant bound and the $\min(\rho_1, \rho_2)$ shared envelope give a q -free majorant, and dominated convergence along $q \rightarrow 0^+$ yields $q^{-r}(J_s^1 - J_s^2) \rightarrow -\delta_r \int u^{s+2k+r} e^{-au^{2k}}$.
- **JetRecovery** (PR #56): the quotient-difference lemma turns this into the normalized covariance limit; at $s = 2k + r$ the covariance is the stage-1 variance, forcing $\delta_r = 0$; strong induction and the scaling-identity transfer give `polynomialJet_recovery`.

With the pure-monomial recovery of (k, a) from the earlier recovery thread, every coefficient of a finite-jet potential is pinned by moment asymptotics, each at its own graded order — the note’s §7.4(a), now marked at its covariance-pairing sentence.

2 The two consults

Both consults contributed mathematics, not just tactics, and both survived contact with the numerics executed before formalisation.

The *envelope consult* rejected the drafted “even-positive-top” integrability condition with a counterexample: $P(y) = 1 - 3y + y^2$ has positive even top yet $P(1) = -1$, so the Gibbs weight keeps a remote peak $e^{q^{-2k}}$ at $u \sim 1/q$ — integrable at every q , never close to the reference measure. Since $L_q(u) = u^{2k}P(qu)$ and qu ranges over all of $\mathbb{R}$ regardless of q , no smallness of q can substitute for profile positivity. This reshaped the programme’s foundations before a line of Lean depended on them.

The *route consult* replaced its own earlier plan. The original stages 3–5 built the full weighted exponential expansion (multi-index coefficient polynomials, triangular decomposition, quotient recurrences). The challenge observation — recovery is a comparison statement, so each rung needs only the *first-order difference* of two Gibbs weights — was endorsed as “the shortest route”, with two load-bearing corrections: the secant bound $|e^{-x} - e^{-y}| \leq |x - y| \max(e^{-x}, e^{-y})$ in place of a Taylor bound whose $e^{|D|}$ factor no envelope controls, and the observation that $\min(\rho_1, \rho_2)$ envelopes the entire segment between the two potentials. The eliminated machinery would only be needed if explicit expansion coefficients were wanted for their own sake.

3 The incident

PR #55 merged an uncompiled file. Its tide never added the file to `Laplace.lean`’s imports, so `lake build` succeeded vacuously, the audit script (which greps text) saw nothing, and CI passed for the same reason. The tide’s log and retrospective claimed a clean build with zero warnings; the claim was false, discovered only when the next tide imported the file and hit fifteen ordinary first-build errors. All were repaired in PR #56; the mathematics was untouched; the false retrospective now carries an erratum. Two durable outputs: the repository rule that a build result may be reported only after verifying the import line (and, for files compiled in the current worktree, the fresh `.olean`), and a transitive import-closure audit which confirmed no other file in the repository shares the exposure. The general lesson for the tide loop: gates fail open exactly when a step is forgotten, so the gate’s own preconditions belong in the close-out checklist.

4 Process observations

Six tide-sized steps against the consult’s original eight-stage plan, with the reroute cutting the middle three stages to two. The stacked-worktree chain (#55’s branch carrying #56’s tide) meant the incident was caught one tide after introduction rather than at some distant integration point — linear chains double as an integration test of the previous link. The numerical checks continued to pay: the response-slope check validated the covariance structure before the route consult returned, and the machine-precision check of the exact scaling identity was effectively free.

5 What the next programme inherits

The normalized-series and quotient-difference handling here was named by the scoping consult as the de-risking prerequisite for the general-order nondegenerate induction (germbij Theorem 3.1 at all orders); the multivariate first rung remains the other open commission, strategically valuable as the Susceptibility Primer’s missing Stage 3. Both need fresh scoping consults. The weighted-jet

machinery itself is reusable: the profile envelope and the pairwise difference limit are stated for arbitrary finite jets, not just the recovery use case.